MODULE 1 PROMPT ENGINEERING

2 API INTEGRATION

SESSION 7: CONVERSATIONAL INTERFACES.

WITH LLMs.

Text Generation

[- Who directed Inception?

Christopher Nolan**

— What other movies has he directed?]

I'm sorry. I can't answer that

① Memory.
② Context
③ Role
④ Behavioral control.

Text generation to Chat completion

- Memory
- Roles.
- behavior consistency.

Roles.

→ User - the one sending the question / human

→ Assistant - the model's response / AI

→ System - contains the settings.

remember

Person 1 — Person 2 — Person 1 — Person 2 ....

purpose, perspective.

System prompt in chat

- Setting the initial mindset:
  reasoning, tone, formatting
- Filters the conversation.
- Drives the behavior.
  - Expertise
  - tone / Brand
  - Structure of the output
  - Sets the expection of the Conversation.

How is an LLM actually chatting?

The cat sat on a _____

["the", "cat", "sat", "on", "a"] → LLM.

mat → 0.65
sofa → 0.20
floor → 0.10
} P > 0.9

Sayan → 0.01
banana → 0.001
eleplant → 0.0001
priyanka → 0.00001

Greedy.
Random Sampling.
topk. K = 5
top P P = 90 / Nucleus sampling.

What happens when we chat

① Message gets wrapped in a Role based object

My name is Reetesh

{ "role" : "user" , "content" : "My name is Reetesh" }

→ { "role" : "system" , "content" : "You are a helpful assistant" }

② Flattening System : You are a helpful assistant

User : ....

③ tokenized.

My → 2048 , Name → 1132 , is → 123 Reetesh → 404864.

④ LLM predicts token by token ⑤ token → text.

⑥ text → object ⑦ Display / return.

Memory is Prompt Engineering magic.

Messages : System, User, Assistant/Model.

Context window limits how far behind model can see.

Gemini — 1M.

50K limit code.

Context Management

- Summarization. = Compress + Save.
- Truncation. = Drop irrelevant.
- Prioritization. = Focus on high value.